LightCtrl: Retinex-Inspired Disentangled Control for Low-Light Image Enhancement

Annicet Razafindratovolahy, Yunbo Rao*, *Member, IEEE*, Shaoning Zeng, *Member, IEEE*, Mona Afanga, Ssemwogerere Rajab, Todizara Andrianajaina

***Abstract*—Low-light image enhancement (LLIE) remains fundamentally challenging due to the strong entanglement between illumination, noise, and scene content, which leads to noise amplification, color distortion, and unstable over-enhancement under severe illumination degradation. Most existing approaches follow a reconstruction paradigm, either by directly mapping low-light inputs to enhanced images or by explicitly estimating illumination and reflectance components. However, because illumination is spatially varying, indirectly observable, and tightly coupled with noise and reflectance, reconstruction errors propagate directly to the output, resulting in limited robustness and poor generalization under real-world conditions. To address this limitation, we introduce a control-based formulation of LLIE, in which illumination is not treated as a reconstruction target but as a control signal that governs enhancement dynamics, enabling a principled separation between what to enhance (content) and how much to enhance (illumination). Based on this formulation, we propose LightCtrl, a structured framework that combines bounded global luminance correction in HSV space with illumination-guided adaptive refinement via disentangled feature representations. At the core of the framework is a unified Control Sensitivity Operator (CSO), a bounded transformation that applies controlled incremental updates and is consistently reused across both image and feature domains to ensure stable, stage-invariant enhancement while preventing saturation and noise amplification. Extensive experiments across multiple datasets demonstrate that the proposed method improves PSNR by 51.4% and reduces NIQE by 50.1% over representative Retinex-based baselines, while maintaining strong perceptual quality and consistent cross-dataset performance with a single fixed model.**

***Index Terms*—Image enhancement, Image processing, Image restoration, Image denoising, Image filtering.**

I. Introduction

DESPITE significant progress in Low-Light Image Enhancement (LLIE), achieving reliable brightness correction without introducing noise amplification or color artifacts remains challenging in real-world scenarios. In low-light conditions, illumination, noise, and scene content are tightly entangled, making enhancement inherently unstable. Many existing methods apply uniform or coupled transformations, which can amplify noise, distort colors, or wash out highlights [1], [2]. This mismatch between where enhancement is needed and how it is applied is a key source of instability.

Recent learning-based approaches attempt to address these issues by estimating illumination maps or learning direct end-to-end mappings from low-light to normal-light images [1], [3], [4]. While effective under controlled conditions, these supervised methods largely follow a *reconstruction paradigm*, in which illumination or image intensities are directly predicted. However, illumination is spatially varying, indirectly observable, and strongly coupled with reflectance and noise [5], making its estimation inherently ill-posed. As a result, estimation errors propagate to the output, leading to over-enhancement, halo artifacts, and reduced robustness under distribution shifts [6], [7], and causing models to overfit to specific training distributions. To compensate for this limitation, recent transformer-based and diffusion-based methods [1], [8], [9] often rely on dataset-specific or metric-specific checkpoint selection, reporting results from different model instances across evaluation settings, which obscures true cross-dataset generalization. Moreover, diffusion-based approaches require iterative sampling, increasing computational cost and potentially introducing hallucinated structures. These limitations extend to unsupervised methods [10]–[13], where enhancement strength remains entangled with scene structure, often resulting in under- or over-enhancement [14].

These observations suggest that directly reconstructing image intensities may be fundamentally misaligned with the nature of LLIE. Instead, illumination can be treated as a *control signal* that regulates enhancement behavior. From this perspective, LLIE can be formulated as a *controlled signal transformation problem*, where enhancement is achieved through stable, bounded updates rather than explicit reconstruction. This enables a principled separation between *what* to enhance (content) and *how much* to enhance (illumination), improving stability and robustness.

Based on this formulation, we propose *LightCtrl*, a control-driven LLIE framework without paired normal-light supervision. The method separates global luminance correction from illumination-guided adaptive refinement. Inspired by Retinex theory, LightCtrl disentangles noise, content, and illumination representations; however, unlike reconstruction-based approaches, these representations are used to *control* enhancement behavior rather than to synthesize image intensities. Specifically, a denoised and color-stable base image is first constructed via bounded luminance enhancement in HSV

*Corresponding authors: Yunbo Rao

A. Razafindratovolahy, Y. Rao, Shaoning Zeng, M. Afanga, S. Rajab are with the University of Electronic Science and Technology of China, Chengdu 611731, China (e-mails: annicet.razafindratovolahy@std.uestc.edu.cn, raoyb@uestc.edu.cn, zeng@csj.uestc.edu.cn, m.afanga@alu.uestc.edu.cn, srajab@std.uestc.edu.cn).

T. Andrianajaina is with the University of Antsiranana, Antsiranana 201, Madagascar (e-mails: todizara.andrianajaina@gmail.com).

This research was supported by the Zhejiang Provincial Natural Science Foundation of China (Grant No. LY23F020025), the Science and Technology Program of Huzhou (Grant No. 2023GZ42), the Science and Technology Project of Sichuan (Grant Nos. 2023ZHCG0005, 2023ZHCG0008), States Key Laboratory of Air Traffic Management System and Technology (Grant No: SKLATM202202), the Chengdu Science and Technology Project (Grant No: 2022-YF05-00068-SN), and Innovation Ability Construction Project for Sichuan Provincial Engineering Research Center of Communication Technology for Intelligent IoT (2303-510109-04-03-318020).

space, followed by a feature-level refinement that adaptively modulates enhancement based on illumination and content.

At the core of the framework is a unified *Control Sensitivity Operator (CSO)*, which defines a bounded incremental update controlled by learned signals. The same operator is reused across stages, enabling consistent and stable enhancement in both image and feature domains. By treating illumination as a control variable rather than a reconstruction target, the proposed design suppresses noise amplification and supports progressive refinement under severe low-light conditions.

The main contributions of this work are summarized as follows:

- We introduce a *control-based formulation of LLIE*, shifting from reconstruction to controlled, bounded signal transformation.
- We propose a unified CSO that enables stable, progressive, and range-bounded enhancement across image and feature domains.
- We show that *illumination–content disentanglement is more effective for control than reconstruction*, improving robustness and generalization.
- We achieve significant performance gains, improving PSNR by 51.4% and reducing NIQE by 50.1% over representative Retinex-based methods, while maintaining consistent cross-dataset performance with lower computational complexity than diffusion-based approaches.

II. Related Work

A. Illumination-Aware Decomposition and Control

Illumination-aware modeling is a central paradigm in LLIE, motivated by the observation that visibility degradation is primarily governed by illumination rather than scene structure. Classical Retinex theory decomposes images into illumination and reflectance components, and recent learning-based methods extend this idea to better handle noise, uneven lighting, and color distortion [15]–[18].

Despite these advances, most approaches rely on estimating illumination as an intermediate representation for reconstruction. Due to the intrinsic coupling between illumination, reflectance, and noise, such estimation is inherently unstable, and errors introduced at this stage tend to propagate to the final output, resulting in over-enhancement and limited robustness.

Beyond explicit decomposition, some methods treat illumination as an adaptive factor to stabilize enhancement across diverse conditions or support downstream tasks [19]–[21]. Optimization-inspired unfolding methods improve interpretability by embedding decomposition objectives into multi-stage networks [5], while physical priors enable zero-reference enhancement through illumination-invariant representations [22]. However, these approaches remain tied to reconstruction-driven formulations and inherit sensitivity to illumination estimation.

In contrast, our method treats illumination as a control signal rather than a reconstruction target, reducing sensitivity to estimation errors and enabling more stable enhancement.

B. Constraint-Driven Enhancement

Given the ill-posed nature of LLIE, many methods introduce constraints to regulate enhancement and suppress artifacts, typically through loss design, physical priors, decomposition objectives, or generative regularization.

Reference-free and self-supervised approaches constrain the solution space using latent quotient estimation or parametric curve formulations, avoiding explicit supervision [23]–[25], while physically motivated constraints jointly stabilize illumination correction and noise suppression [21]. Although effective, these methods regulate enhancement indirectly and remain sensitive to loss design.

Other approaches explicitly model illumination–reflectance interactions. Retinex-inspired models incorporate decomposition objectives, while unfolding-based methods enforce structural priors through iterative optimization [5], [20]. Despite improved interpretability, their performance depends on accurate decomposition.

More recent generative frameworks impose constraints through the generation process itself. Diffusion-based methods guide refinement using anchoring mechanisms or prior-driven attention [18], [26], while physical priors enable zero-reference enhancement via illumination-invariant representations [22]. However, these approaches often involve high computational cost and limited controllability.

In contrast, LightCtrl constrains enhancement at the level of transformation dynamics, directly controlling enhancement through illumination-guided modulation within a bounded operator, enabling stable and predictable behavior.

C. Iterative and Operator-Based Enhancement Methods

Many LLIE methods formulate enhancement as an iterative refinement process, where improvements are achieved through repeated application of an update operator, enabling finer control over enhancement dynamics under complex degradations.

Early approaches rely on illumination estimation followed by structured refinement, while Retinex-based methods formulate enhancement as alternating optimization between illumination and reflectance [27], [28]. Learning-based extensions unroll iterative solvers into multi-stage networks [5], [29], and decomposition-driven pipelines further separate illumination adjustment from noise and color correction [30].

Iterative refinement is also achieved through feedback mechanisms. Back-projection architectures iteratively correct residuals, while curve-based methods enforce monotonic and bounded transformations [31], [32]. More recent approaches explore implicit operator learning through self-regression and normalization [25].

Generative models introduce a different iterative paradigm based on sampling trajectories. Diffusion-based methods progressively refine images while incorporating priors to maintain fidelity and suppress artifacts [18], [26], with extensions to more complex settings such as low-light light-field restoration [33].

These works highlight the importance of operator design in controlling enhancement behavior. However, most approaches rely on reconstruction-driven updates or lack explicit control

TABLE I
CONCEPTUAL COMPARISON OF LOW-LIGHT IMAGE ENHANCEMENT PARADIGMS

Method Type	Representative Works	Core Strategy	Illumination Role	Key Limitation
Retinex-based	RetinexNet, KinD++, Retinexformer	Decomposition into illumination and reflectance	Explicitly estimated and reconstructed	Error propagation from illumination estimation
Curve-based	Zero-DCE++	Learn parametric intensity transformation curves	Implicit modulation	Limited adaptability to complex scenes
Unfolding-based	Optimization-inspired methods	Iterative approximation of optimization process	Estimated as intermediate variable	Sensitive to modeling assumptions
Diffusion-based	LLDDiffusion and related models	Progressive stochastic denoising and generation	Implicit latent control	High computational cost and risk of hallucination
Proposed (LightCtrl)	—	**Control-driven transformation**	**Control signal (not reconstructed)**	Requires appropriate control design

over enhancement dynamics. In contrast, LightCtrl introduces a unified bounded operator explicitly modulated by illumination, enabling consistent and stage-invariant refinement. Methods based on strong supervision, such as RAW-guided approaches, share staged designs but depend on explicit ground-truth signals [34].

To further clarify the distinction between existing paradigms and the proposed approach, we summarize key conceptual differences in Table I.

III. METHODOLOGY

A. Method Overview

An overview of the proposed framework is shown in Fig. 1. Given a low-light input image, a shared encoder first extracts a compact representation that captures information relevant to noise, illumination, and scene content. From this shared representation, lightweight decoders estimate auxiliary signals corresponding to these factors. These signals are not used to reconstruct image components explicitly; instead, they serve as control inputs that guide enhancement at different stages of the pipeline.

At the core of the framework is a single enhancement operator, referred to as the CSO. It defines a bounded incremental update whose magnitude can be modulated by external control signals. As illustrated in Fig. 1, the same operator is reused across multiple stages of the method, while its role varies according to the semantic meaning of its input.

The enhancement process proceeds in two stages. First, a global exposure baseline is constructed by applying CSO iteratively to the luminance channel in HSV space, producing a color-consistent image with corrected overall brightness. Second, spatially adaptive refinement is performed using illumination- and content-related signals derived from the shared encoder.

By combining a shared representation with a unified enhancement operator, the proposed method achieves stable global correction and adaptive local refinement within a coherent and reusable framework.

B. Control Sensitivity Operator

Stage-Invariant Enhancement Requirement. The proposed framework performs enhancement across multiple stages with distinct operational regimes. Specifically, luminance construction in HSV space is carried out through iterative updates, while feature interaction is performed in a single forward pass, and final image refinement again follows an iterative process. Despite these differences, all stages are governed by a unified control signal derived from illumination.

A natural question is why a single reusable operator is required across these stages. If different transformation functions are employed, the same control signal would induce inconsistent effects depending on where it is applied, breaking the coherence of the enhancement process. In particular, the HSV stage constructs a base illumination through repeated application, while the final refinement stage further adjusts the image through another iterative process, and the feature interaction stage applies the transformation only once. Without a shared operator, these transformations would belong to different functional families, making the effect of the control signal stage-dependent and leading to instability and loss of interpretability. Therefore, a stage-invariant operator is required to ensure that enhancement behaves as a consistent, control-driven transformation across all stages of the pipeline.

Axiomatic Formulation. We define an operator $F(x; r)$ over normalized inputs $x \in [0, 1]$ with control variable $r \geq 0$, satisfying the following properties:

Axiom 1 (Control–Shape Invariance):

$$F(x; r) = x + \rho(r)\, g(x), \tag{1}$$

where $g(x)$ is independent of r.

Axiom 2 (Self-Limiting Behavior):

$$g(x) = (1 - x)\, h(x), \tag{2}$$

ensuring bounded updates as $x \to 1$.

Axiom 3 (Mixed-Mode Consistency):

$$F^{(K)}(x; r) \approx F(x; \psi_K(r)), \tag{3}$$

where ψ_K is a monotonic mapping. This condition is required because the operator is applied iteratively in both the HSV luminance construction and final refinement stages, while it is applied only once in the feature interaction stage.

Axiom 4 (Non-Degenerate Low-Signal Response):

$$h(x) \sim x \quad \text{as } x \to 0. \tag{4}$$

From Axioms 1 and 2, the operator admits the general form:

$$F(x; r) = x + \rho(r)\,(1 - x)\, h(x). \tag{5}$$

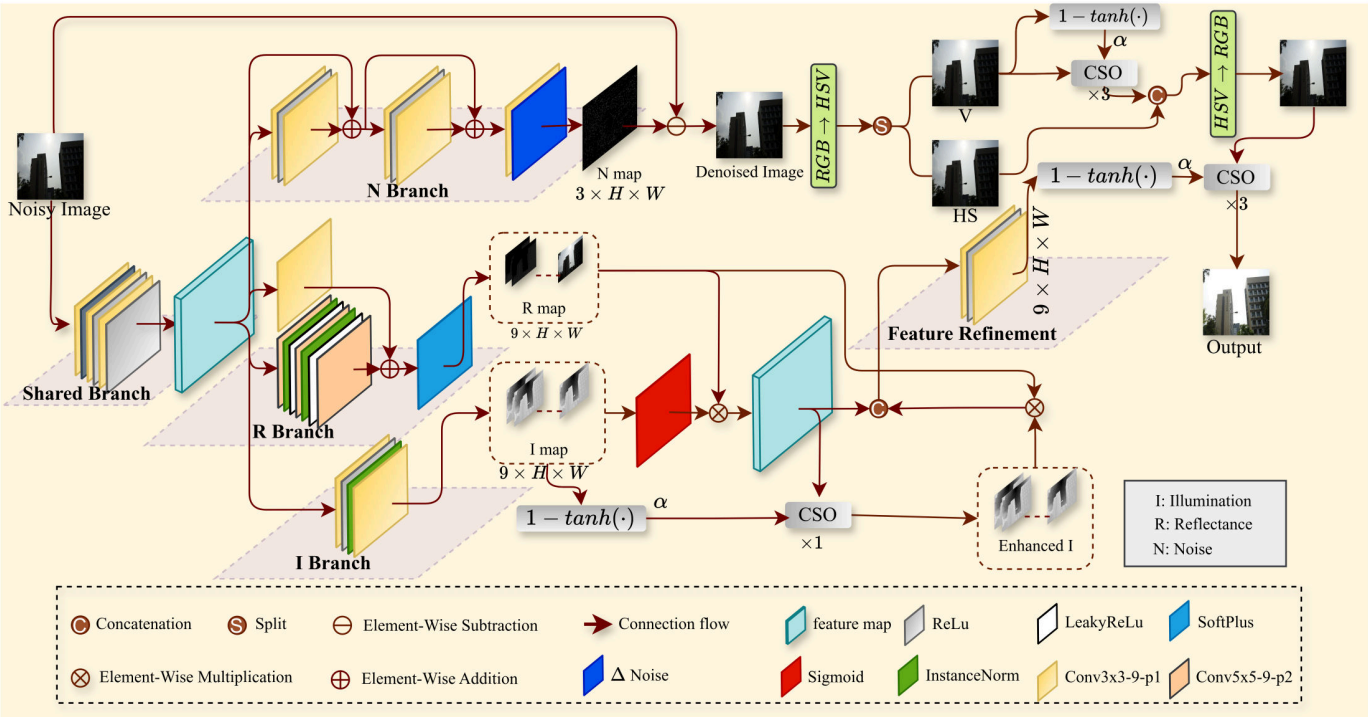

Fig. 1. Overview of the proposed LLIE framework. The architecture decomposes enhancement into noise estimation, reflectance modeling, and illumination-guided modulation.

a) Necessity of Higher-Order Sensitivity: Consider the lowest-order admissible form $h(x) = c_1 x$, yielding:

$$F(x; r) = x + \lambda(r)\,(1 - x)x. \tag{6}$$

This formulation defines a quadratic update. However, this class of operators is not closed under composition. Specifically, repeated application produces higher-degree polynomials that cannot be expressed within the same functional form using a single reparameterized control variable. As a result, such operators violate Axiom 3 and fail to maintain consistency between iterative and single-step regimes.

To satisfy mixed-mode consistency, the response function must incorporate higher-order signal sensitivity:

$$h(x) = c_1 x + c_3 x^3 + O(x^5). \tag{7}$$

The cubic term introduces the minimal additional degree of freedom required to approximate compositional behavior while preserving smoothness and boundedness.

Control Sensitivity Operator (CSO). We adopt the lowest-order realization:

$$h(x) = \frac{1}{2}\left(x + \frac{x^3}{6}\right), \tag{8}$$

leading to the CSO:

$$\mathrm{CSO}(x; r) = x + \frac{1 - x}{2}\left(rx + \frac{rx^3}{6}\right). \tag{9}$$

This operator satisfies all axioms and represents the minimal solution enabling stage-invariant enhancement across both iterative and single-step regimes. This choice corresponds to a truncated odd-order expansion ensuring smooth compositional approximation.

Cross-Stage Behavior. The CSO is applied under three distinct regimes within the proposed framework:

- **HSV luminance construction:** CSO is applied iteratively for K steps, producing a progressive and bounded accumulation of global brightness.
- **Feature interaction:** CSO is applied once to illumination-conditioned features, enabling controlled modulation without iterative accumulation.
- **Final image refinement:** CSO is applied iteratively for K steps on the base image, allowing spatially adaptive refinement under learned control.

Despite these differing application modes, the operator maintains consistent behavior due to its compositional stability and bounded update structure.

Summary. The proposed CSO is not an empirical design choice, but the minimal solution derived from stage-invariant enhancement requirements. By incorporating higher-order signal sensitivity within a bounded update framework, it enables a unified transformation rule that operates coherently across iterative luminance construction, feature-level modulation, and iterative image refinement.

C. Global Enhancement via the HSV Path

The HSV path constructs a denoised and globally enhanced base image by operating on the luminance component while preserving chromatic consistency. This stage provides an initial exposure correction that serves as the foundation for subsequent refinement.

Fig. 2. Preliminary base enhancement produced by the HSV path with different iteration numbers K. Increasing K progressively improves global exposure. The setting K=3 provides a balanced base enhancement used in our method, while fine-grained detail and local adjustment are handled by the subsequent refinement stage.

Given the input image $x \in \mathbb{R}^{3\times H\times W}$, a shared encoder extracts a latent representation, from which a noise decoder predicts an additive noise residual

$$\hat{N} = D_N(E(x)) \in \mathbb{R}^{3\times H\times W}. \tag{10}$$

The denoised image is obtained as

$$x_{\text{den}} = \text{clip}(x - \hat{N}, 0, 1), \tag{11}$$

which suppresses noise prior to luminance amplification.

The denoised image x_{den} is converted to the HSV color space, and enhancement is applied exclusively to the Value channel v. A luminance-dependent control factor is computed as

$$\alpha_v = 1 - \tanh(v), \tag{12}$$

assigning stronger enhancement control to low-luminance regions while progressively attenuating enhancement in brighter areas. Global exposure is constructed through iterative application of the CSO,

$$v^{(k+1)} = \text{CSO}\left(v^{(k)}; \alpha_v\right), \quad k = 0, \ldots, K-1, \tag{13}$$

with $v^{(0)} = v$.

After K iterations, the enhanced Value channel is combined with the original Hue and Saturation channels and converted back to RGB, producing a base enhanced image x_0. Fig. 2 illustrates the effect of the iterative HSV enhancement for different iteration numbers. In particular, K=3 provides a balanced operating point: fewer iterations (e.g., K=2) lead to insufficient exposure correction, whereas more iterations (e.g., K=4) may produce an overly bright base image that leaves limited headroom for subsequent processing.

However, a balanced base enhancement at K=3 alone is still not sufficient to fully recover fine details and local structures in all scenes. As shown in the ablation study (Sec. IV-A), removing the illumination–reflectance refinement stage results in locally suboptimal enhancement, especially in challenging regions. This motivates the refinement stage introduced in the next section, which leverages illumination- and content-aware signals to further improve local detail while preserving the global exposure established by the HSV path.

D. Illumination–Reflectance Guided Refinement

While the HSV path establishes a globally stable luminance correction by enhancing only the Value channel, it operates independently of scene structure and spatial illumination variation. As a result, global luminance correction alone may be insufficient to adapt enhancement strength to locally varying conditions. To address this limitation, an illumination–reflectance guided refinement path is introduced, which operates in feature space and predicts a spatially varying modulation for the HSV-enhanced base image.

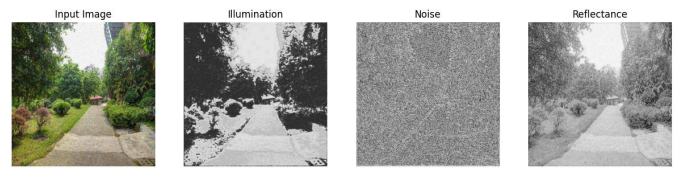

Fig. 3. Visualization of PCA-projected feature maps from the three specialized branches. Each branch exhibits distinct spatial patterns corresponding to noise, lighting, and structural information.

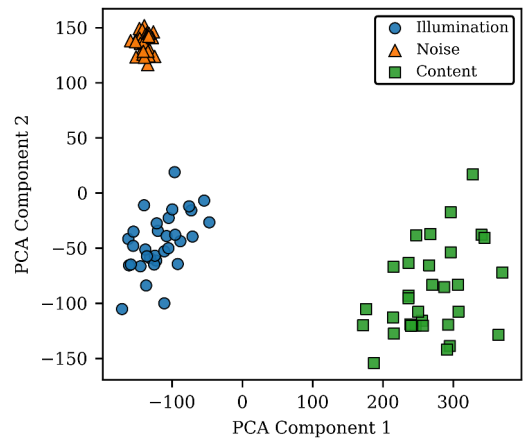

Fig. 4. 2D PCA projection of sampled features from the Noise, Illumination, and Reflectance branches. Each point represents a spatial feature vector projected into a common coordinate space. The clustering indicates effective separation of noise, lighting, and structural information.

Two decoders extract content and illumination features from the shared encoder output:

$$R = D_R(E(x)), \quad I = D_I(E(x)), \tag{14}$$

where $R, I \in \mathbb{R}^{9\times H\times W}$. The reflectance features encode structural information such as edges and spatial layout, while the illumination features represent local lighting conditions indicating where stronger or weaker refinement is required.

Fig. 3 visualizes the outputs of the three specialized branches after projecting the feature maps into RGB space using PCA. The noise branch emphasizes stochastic high-frequency components, the illumination branch highlights smooth intensity variations corresponding to lighting, and the reflectance branch preserves structural details such as edges and object boundaries. These distinct spatial patterns indicate that each branch learns representations aligned with its intended role.

To further analyze the separation of learned feature representations, Fig. 4 shows a two-dimensional PCA projection of sampled features from the noise, illumination, and content branches. Each point corresponds to a spatial feature vector projected into a shared coordinate space.

The resulting distribution shows clear clustering of features from the three branches, indicating that the learned representations occupy distinct subspaces. This separation supports effective disentanglement of noise, illumination, and structural content, enabling illumination features to guide enhancement in a spatially adaptive and controlled manner.

Illumination features are then used in two complementary roles to control refinement behavior. First, illumination is mapped to a spatial gating signal,

$$I_g = \sigma(I), \tag{15}$$

which modulates the content features through element-wise multiplication:

$$F_{\text{ctx}} = R \odot I_g. \tag{16}$$

This conditioning allows refinement strength to vary spatially according to local illumination while preserving the underlying structural representation encoded in R.

Second, the raw illumination features are transformed into a modulation-strength signal,

$$\alpha_I = 1 - \tanh(I), \tag{17}$$

assigning larger values to darker regions where stronger refinement is required. This signal controls a single application of the CSO on the illumination-conditioned content features:

$$\tilde{F}_{\text{ctx}} = \text{CSO}(F_{\text{ctx}}; \alpha_I). \tag{18}$$

This feature-space update serves to produce a bounded, illumination-aware refinement cue. After this adjustment, the updated features are fused with the raw content features. We re-inject R in Eq. 19 to anchor refinement cues to the original structural subspace and prevent illumination-driven feature drift:

$$F_{\text{en}} = R \odot \tilde{F}_{\text{ctx}}. \tag{19}$$

Finally, the original illumination-conditioned features and the refined features are concatenated,

$$G = \text{cat}(F_{\text{ctx}}, F_{\text{en}}), \tag{20}$$

and passed to a lightweight refinement module, which predicts a modulation tensor $\alpha_M \in \mathbb{R}^{9\times H\times W}$:

$$\alpha_M = 1 - \tanh(M), \tag{21}$$

which is applied to the HSV-enhanced base image x_0 through the CSO. The final image refinement follows the same iterative formulation as the HSV path, where CSO is applied for K steps to progressively refine the base image. Fig 5 shows representative examples from the LOL dataset, including the input image, ground truth, final output, and the corresponding refinement activation. Although the base enhancement provides a globally corrected image, the refinement activation highlights spatially varying regions that require further adjustment, such as textured areas and illumination transitions. This observation indicates that the refinement stage operates selectively rather than uniformly, complementing the base enhancement by focusing on regions that are difficult to recover through global luminance correction alone.

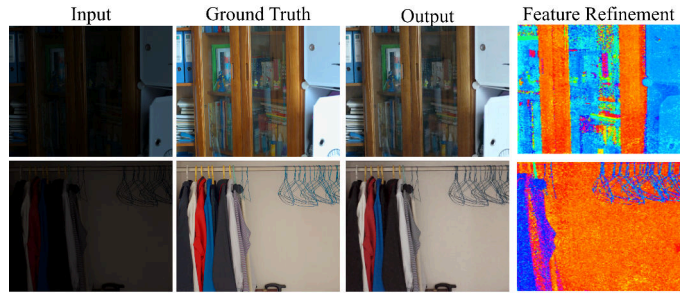

Fig. 5. Representative examples from the LOL dataset showing the input image, ground truth, final enhanced output, and the corresponding refinement activation. The refinement is spatially selective, with stronger responses in textured regions and illumination transitions, complementing the global correction provided by the base enhancement.

Fig. 6. Refinement results obtained from different HSV base enhancements. The refinement stage is applied to base images produced with K=2, K=3, and K=4. A balanced base enhancement (K=3) enables effective refinement, while an under-corrected base (K=2) limits improvement and an over-corrected base (K=4) leads to over-exposure due to further amplification during refinement.

As shown in Fig. 6, the effect of the refinement stage depends on the base enhancement provided by the HSV path. When the base image is under-corrected (K=2), refinement improves local contrast but global brightness remains insufficient. With the balanced setting (K=3), refinement effectively enhances local details while preserving stable global exposure. In contrast, when the base image is already over-enhanced (K=4), the refinement stage further amplifies pixel intensities through the CSO, leading to over-exposure. This observation highlights the importance of using a moderately corrected base image for stable refinement.

E. Objective function

The network is trained using a composite objective

$$\begin{aligned}\mathcal{L}_{\text{total}} = \lambda_{\text{ucl}}\mathcal{L}_{\text{UCL}} + \lambda_{\text{epl}}\mathcal{L}_{\text{EPL}} + \lambda_{\text{noise}}\mathcal{L}_{\text{noise}} \\ + \lambda_{\text{indep}}\mathcal{L}_{\text{indep}} + \lambda_{\text{spa}}\mathcal{L}_{\text{spa}} + \lambda_{\text{tv}}\mathcal{L}_{\text{tv}} + \lambda_{\text{col}}\mathcal{L}_{\text{col}}.\end{aligned} \tag{22}$$

This objective regularizes the enhancement behavior induced by the CSO by constraining exposure, structure, feature disentanglement, and modulation smoothness.

To guide spatially adaptive enhancement, we use the Underexposure Correction Loss (UCL) and Exposure Preservation Loss (EPL), following the original implementation in [35].

$$\mathcal{L}_{\text{UCL}} = \frac{1}{P}\sum_{p=1}^{P} \sigma\left(k(t-\mu_p^{\mathbf{x}})\right) \cdot \max\left(0, t-\mu_p^{\mathbf{x}_{\text{enh}}}\right)^2 \tag{23}$$

$$\mathcal{L}_{\text{EPL}} = \frac{1}{P}\sum_{p=1}^{P} \sigma\left(k(\mu_p^{\mathbf{x}}-t)\right) \cdot \left(\mu_p^{\mathbf{x}_{\text{enh}}}-\mu_p^{\mathbf{x}}\right)^2 \tag{24}$$

Here, $\mu_p^{\mathbf{x}}$ and $\mu_p^{\mathbf{x}_{\text{enh}}}$ are the mean luminance of patch p in the input and enhanced images, respectively; t is a luminance threshold; k controls sigmoid sharpness; $\sigma(\cdot)$ is the sigmoid function; and P is the total number of patches. UCL promotes brightening in dark regions, while EPL preserves well-exposed areas.

Noise supervision uses synthetic noisy-clean image pairs $(x^{\text{noisy}}, x^{\text{clean}})$:

$$\mathcal{L}_{\text{noise}} = \left\| \hat{N} - (x^{\text{noisy}} - x^{\text{clean}}) \right\|_2^2. \tag{25}$$

Let n, r, and i denote global-average-pooled features from $\hat{N}$, R, and I, respectively, and $\bar{n}$, $\bar{r}$, and $\bar{i}$ their ℓ_2-normalized forms. Feature disentanglement is enforced by

$$\mathcal{L}_{\text{indep}} = |\langle \bar{i}, \bar{n} \rangle| + |\langle \bar{i}, \bar{r} \rangle| + |\langle \bar{n}, \bar{r} \rangle|. \tag{26}$$

To preserve local structural relationships, we introduce a spatial consistency loss using a neighborhood set D:

$$\mathcal{L}_{\text{spa}} = \frac{1}{HW|D|} \sum_{i,j} \sum_{d \in D} \left\| (x_{i,j}^{\text{enh}} - x_{i+d}^{\text{enh}}) - (x_{i,j} - x_{i+d}) \right\|_2^2. \tag{27}$$

We further regularize the modulation map α using total variation [36]:

$$\mathcal{L}_{\text{tv}} = \frac{1}{HW} \sum_{i,j} \left(\|\nabla_h \alpha_{i,j}\|_2^2 + \|\nabla_v \alpha_{i,j}\|_2^2 \right). \tag{28}$$

A color constancy loss is used to reduce color bias, inspired by classical color constancy principles [37]:

$$\mathcal{L}_{\text{col}} = (\mu_r - \mu_g)^4 + (\mu_r - \mu_b)^4 + (\mu_g - \mu_b)^4, \tag{29}$$

where μ_r, μ_g, and μ_b denote mean RGB intensities of x^{enh}.

IV. Experiments and results

Our model is trained on a combined training set used in prior works [32], with synthetic Gaussian noise ($\sigma = 0.1$) added to simulate sensor noise. We implement our framework using *PyTorch*, train with a *batch size of 32*, and use the *Adam optimizer* with an initial learning rate of 1×10^{-5}. Training is conducted on an *NVIDIA RTX 3090 GPU*. We fix λ_{exp}=10, λ_{noise}=1, λ_{indep}=0.1, λ_{spa}=5, λ_{tv}=0.2, and λ_{col}=0.05 for all experiments.

Table II reports quantitative comparisons on both paired and unpaired low-light benchmarks. On paired datasets, LightCtrl achieves competitive PSNR and SSIM, indicating effective recovery of global luminance while preserving structural information. On unpaired benchmarks, LightCtrl consistently attains strong perceptual scores (NIQE and BRISQUE), reflecting visually natural enhancement under real-world conditions. Importantly, these results are obtained using a single fixed checkpoint and with lower computational cost than recent diffusion-based or transformer-based approaches, highlighting the efficiency of the proposed framework.

We observe that some recent transformer-based and diffusion-based methods exhibit lower performance under our evaluation protocol. One possible reason is that we use a single checkpoint across all datasets and evaluation metrics, without dataset-specific or metric-specific tuning. In contrast, the performance reported in original works may benefit from checkpoint selection tailored to specific datasets or metrics. As a result, using a unified checkpoint can reveal differences in cross-dataset generalization and robustness.

The qualitative comparisons in Fig. 7 and Fig. 8 provide representative visual correspondence to the quantitative trends observed in Table II. While several competing methods improve brightness effectively, they often exhibit characteristic artifacts such as color bias, local over-enhancement, or unstable highlights, particularly in regions with strong illumination contrast. In contrast, LightCtrl produces balanced enhancement by lifting under-exposed regions while maintaining chromatic stability and avoiding excessive amplification. These visual observations explain why LightCtrl achieves favorable perceptual scores in Table II while remaining computationally efficient, as it avoids heavy illumination reconstruction or costly iterative sampling.

For further analysis, we select representative baselines from different methodological categories: Retinex-Net as a classical illumination reconstruction approach, LLFormer as a recent learning-based enhancement method, and Zero-DCE++ as a widely used unsupervised curve-based model. Fig. 9 reports PSNR and NIQE across different input illumination levels. As illumination increases, all methods exhibit improved performance. Notably, LightCtrl achieves larger gains in severely low-light bins, indicating more effective enhancement under challenging conditions. In contrast, Retinex-Net and LLFormer show reduced stability in darker inputs, while Zero-DCE++ remains conservative but under-enhances. Performance gaps gradually narrow in brighter bins, where enhancement becomes less critical.

A. Ablation Study

The ablation results in Table III show that each component contributes meaningfully to the final performance. Removing illumination-guided modulation degrades both perceptual and distortion metrics, confirming its importance for adaptive enhancement. Disabling iterative refinement consistently reduces visual quality, indicating that progressive updates are necessary to avoid under- or over-exposure. Performing enhancement directly in RGB space instead of HSV further deteriorates perceptual scores due to increased color artifacts. Overall, these results demonstrate that the proposed components are complementary and that the final configuration achieves a balanced trade-off between robustness and simplicity.

Fig. 10 provides a visual comparison highlighting the roles of the HSV-based enhancement and the illumination–reflectance (I/R) refinement path. Without the I/R path, enhancement relies solely on HSV-based Value correction, which preserves hue and saturation but often introduces posterization and unnatural tonal transitions in structurally complex regions. Removing the CSO from the I/R path leads to less stable refinement, resulting in inconsistent enhancement across regions. In contrast, the full model effectively suppresses HSV-induced artifacts and produces more natural and coherent results.

We visualize only these two ablations because they correspond to the most perceptually evident failure modes: the

TABLE II

QUANTITATIVE COMPARISON ON PAIRED (LOL-V1, LOL-V2-REAL, LOL-V2-SYN, MIT-ADOBE FIVEK) AND UNPAIRED (DICM, LIME, MEF, NPE, VV) DATASETS. FOR PAIRED DATASETS, PSNR↑ AND SSIM↑ ARE REPORTED, WHILE FOR UNPAIRED DATASETS, NIQE↓ AND BRISQUE↓ ARE USED. THE LAST COLUMN REPORTS MODEL EFFICIENCY IN TERMS OF THE NUMBER OF PARAMETERS (IN MILLIONS, PARAM.(M)↓). BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND UNDERLINE, RESPECTIVELY. METHODS ARE CATEGORIZED AS TRADITIONAL (T), SUPERVISED (S), AND UNSUPERVISED/ZERO-SHOT (U).

Type	Method	LOL-v1 [38]		LOL-v2-real [39]		LOL-v2-syn [39]		MIT FiveK		DICM [40]		LIME [41]		MEF [42]		NPE [43]		VV[1]		Complexity
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	NIQE↓	BRISQUE↓	NIQE↓	BRISQUE↓	NIQE↓	BRISQUE↓	NIQE↓	BRISQUE↓	NIQE↓	BRISQUE↓	Param.(M)↓
T	Razafi et al. [44]	19.45	0.78	20.08	**0.85**	17.98	0.80	22.63	0.84	**3.52**	21.50	3.82	15.03	3.65	17.18	4.74	17.87	4.09	7.86	–
	TOVO [45]	20.47	0.80	20.40	0.79	20.60	0.81	22.09	0.82	3.53	**14.81**	3.73	15.20	3.78	16.02	4.09	17.91	2.68	7.28	–
	DUAL [46]	15.95	0.68	14.99	0.48	17.34	0.77	15.39	0.59	3.75	17.68	**3.65**	21.55	3.70	24.03	4.86	19.35	3.67	13.58	–
S	RetinexNet [47]	16.21	0.69	15.62	0.65	19.40	0.62	21.86	0.84	4.45	29.34	4.75	30.50	**3.46**	21.34	4.85	31.00	4.80	31.20	0.84
	KinD++ [48]	17.80	0.73	18.50	0.79	20.50	0.89	21.94	0.87	3.78	22.91	3.77	22.79	3.88	30.31	4.28	**16.85**	3.67	9.43	8.27
	Retinexformer [49]	17.82	0.76	18.28	0.79	19.56	0.81	18.43	0.78	4.06	24.73	3.70	23.12	4.01	29.51	4.35	20.08	3.50	6.80	1.61
	LLFormer [9]	24.63	0.87	17.50	0.78	18.41	0.78	20.14	0.82	4.75	30.55	4.98	35.54	4.10	27.56	5.17	31.72	4.42	31.98	13.15
	LLDDiffusion [1]	23.39	**0.89**	20.20	0.78	19.44	0.78	20.30	0.89	6.10	19.77	4.98	13.00	4.10	27.56	5.80	31.80	4.62	7.33	226.18
	PairLIE [50]	16.03	0.50	19.86	0.80	19.07	0.84	20.40	0.87	4.00	31.68	4.60	32.5	4.17	29.54	4.65	23.6	5.20	33.8	0.34
	LIEDNet-L [3]	18.10	0.76	18.33	0.77	18.30	0.83	17.93	0.76	4.90	25.5	3.80	16.9	3.71	16.40	4.85	25.00	3.82	25.20	17.79
U	DFF-Net [51]	19.00	0.79	19.10	0.84	20.35	0.80	24.66	0.77	4.45	19.00	3.91	16.2	3.90	29.60	4.00	25.3	5.95	19.5	5.15
	CSBNet [52]	20.90	0.83	**21.84**	0.84	**22.14**	0.82	19.75	0.81	4.10	21.0	4.00	20.5	3.95	19.9	**3.05**	20.7	3.00	15.80	2.72
	EnlightenGAN [53]	17.08	0.69	17.80	0.72	19.50	0.60	17.80	0.72	4.10	27.5	4.00	17.00	3.90	26.4	4.23	17.32	4.02	21.20	8.64
	Zero-DCE++ [32]	17.01	0.68	18.50	0.60	17.80	0.78	19.00	0.80	3.66	17.45	3.66	16.09	4.20	28.3	4.35	29.10	2.67	9.43	**0.011**
	Ours	**24.95**	0.88	19.93	0.78	20.90	**0.90**	**25.42**	**0.91**	3.55	16.45	**3.65**	**12.77**	3.71	**15.10**	4.00	17.06	**2.64**	**6.29**	0.013

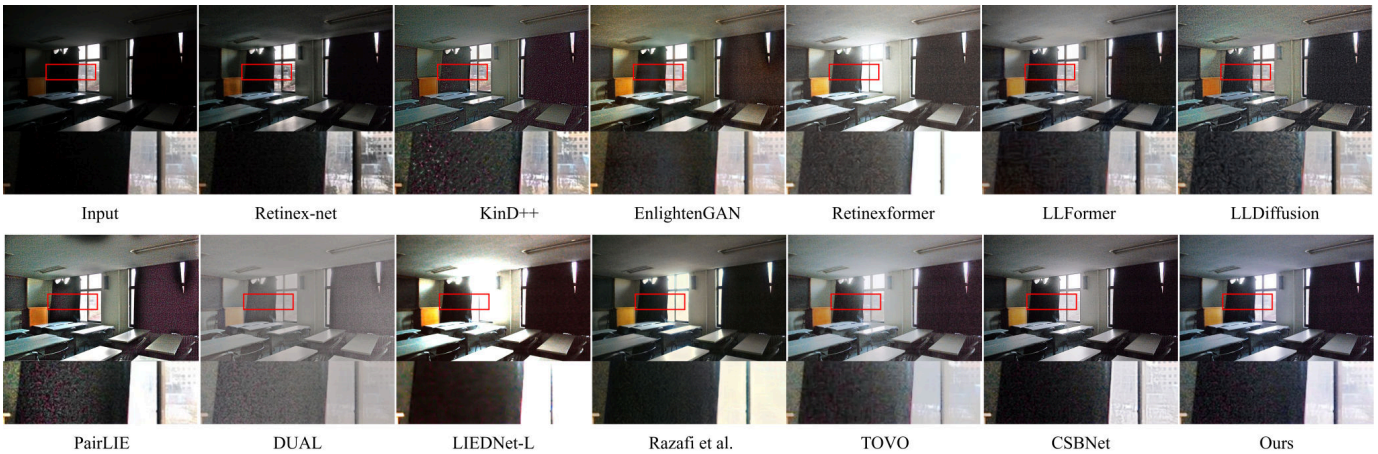

Fig. 7. Qualitative comparison on representative low-light images. LightCtrl produces balanced enhancement results by refining a color-stable HSV-based correction through illumination–content guided modulation, avoiding excessive brightening and visually unstable artifacts.

TABLE III

COMPONENT-WISE ABLATION STUDY. EACH ROW SHOWS THE PRESENCE (✓) OR ABSENCE (✗) OF MAJOR MODULES AND THE NUMBER OF CSO ITERATIONS. METRICS ARE EVALUATED ON THE LOL DATASET. NOTATION FOLLOWS TABLE II.

Noise Branch	I/R path	HSV Enh.	Refine	CSO Iter.	PSNR ↑	SSIM ↑
✓	✓	✓	✓	3	**24.95**	**0.88**
✗	✓	✓	✓	3	23.0	0.85
✓	✗	✓	✓	3	21.8	0.82
✓	✓	✗	✓	3	22.9	0.84
✓	✓	RGB	✓	3	23.5	0.86
✓	✓	✓	✗	3	23.2	0.85
✗	✓	✗	✗	3	21.5	0.80
✓	✓	✓	✓	1	23.8	0.86

absence of feature-based refinement and the absence of CSO. Other ablations primarily affect quantitative performance and are therefore reported only in Table III.

B. Structural Analysis

The gradient responses in Fig. 11 highlight that conventional enhancement (e.g., Retinex) tends to amplify local contrast without preserving global structural coherence, leading to fragmented and inconsistent edge patterns. In contrast, our method maintains both local detail and long-range structural continuity, indicating that the enhancement process is guided by a more stable and structure-aware transformation rather than purely local intensity amplification.

V. LIMITATIONS AND FUTURE WORK

While LightCtrl demonstrates strong robustness and stability across diverse low-light conditions, it relies on a fixed iteration schedule in the HSV-based global enhancement stage. Specifically, the number of iterative updates (K) is empirically chosen to balance sufficient exposure correction and preservation of headroom for subsequent refinement. Although this fixed setting generalizes well across datasets, it may not be optimal for all lighting distributions. Scenes with extreme underexposure or near-normal illumination could benefit from a more adaptive control of iteration depth or exposure strength.

Future work may explore adaptive or learnable control schedules that dynamically adjust the number or strength of updates based on image content. Such extensions could further improve flexibility while maintaining the bounded and stable behavior of the proposed control mechanism. Additionally, extending the control formulation to incorporate more context-aware or scene-dependent priors may enhance performance under more challenging or diverse real-world conditions.

More broadly, the proposed CSO is not inherently restricted to low-light enhancement. As a bounded, control-driven trans-

Fig. 8. Multi-scene qualitative comparison on diverse low-light images. Representative results from different methods are shown across outdoor, architectural, and indoor scenes. While competing approaches exhibit scene-dependent failure modes such as under-enhancement, over-exposure, or color distortion, LightCtrl produces visually stable results with balanced brightness and preserved structural details across all scenes.

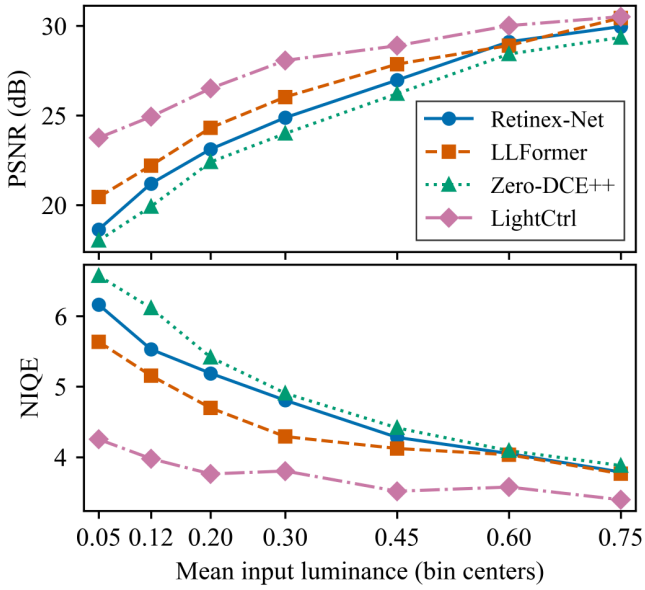

Fig. 9. PSNR and NIQE across input-illumination bins.

Fig. 10. A visual ablation: removing illumination-guided modulation (w/o illum) and removing the CSO update (w/o CSO).

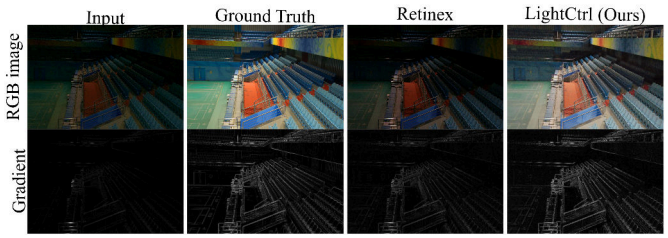

Fig. 11. Comparison of structural restoration using gradient visualization. While Retinex fails to recover fine details under low-light conditions, our method effectively restores structural information consistent with the ground truth.

formation, it provides a general mechanism for progressive signal adjustment that could be applicable to other enhancement and restoration tasks. Exploring its integration into different domains, such as dehazing or exposure correction, as well as its role in alternative feature representations, constitutes an interesting direction for future research.

VI. Conclusion

In this work, we presented LightCtrl, a control-driven LLIE framework that reformulates enhancement as a bounded and progressive update process rather than a direct reconstruction problem. By treating illumination as a control signal instead of a reconstruction target, the proposed approach enables a principled separation between content and enhancement strength, leading to improved stability under challenging low-light conditions.

The proposed framework combines global luminance correction with illumination-guided adaptive refinement, supported by a unified CSO that enforces consistent and bounded transformations across stages. This design effectively suppresses noise amplification and over-enhancement while preserving structural and chromatic fidelity.

Extensive experiments demonstrate that LightCtrl achieves

strong quantitative and perceptual performance with consistent cross-dataset behavior using a single fixed model. These results highlight the advantages of control-based formulations over reconstruction-driven approaches in terms of robustness, stability, and generalization.

Overall, this work suggests that modeling enhancement as a controlled transformation provides a more reliable and interpretable framework for LLIE, offering a promising direction for future research beyond reconstruction-based paradigms.

References

[1] T. Wang, K. Zhang, Y. Zhang, W. Luo, B. Stenger, T. Lu, T.-K. Kim, and W. Liu, "Lldiffusion: Learning degradation representations in diffusion models for low-light image enhancement," *Pattern Recognition*, vol. 166, p. 111628, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320325002882

[2] S. Du, Y. Wen, M. Zhao, J. Li, Z. Shi, and Y. Liu, "A decoupled framework for low-light image enhancement," *Expert Systems with Applications*, vol. 297, p. 129436, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417425030520

[3] M. Liu, Y. Cui, W. Ren, J. Zhou, and A. C. Knoll, "Liednet: A lightweight network for low-light enhancement and deblurring," *IEEE Transactions on Circuits and Systems for Video Technology*, 2025.

[4] X. Zeng, L. Zhu, W. Yang, H. Leung, S. Wang, and S. Kwong, "Low-light image enhancement via diffusion models with semantic priors of any region," *IEEE Transactions on Circuits and Systems for Video Technology*, pp. 1–1, 2025.

[5] W. Wu, J. Weng, P. Zhang, X. Wang, W. Yang, and J. Jiang, "Interpretable optimization-inspired unfolding network for low-light image enhancement," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 4, pp. 2545–2562, 2025.

[6] H. Zhou, W. Dong, X. Liu, Y. Zhang, G. Zhai, and J. Chen, "Low-light image enhancement via generative perceptual priors," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 10, pp. 10 752–10 760, Apr. 2025. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/33168

[7] L. Ma, D. Jin, N. An, J. Liu, X. Fan, Z. Luo, and R. Liu, "Bilevel fast scene adaptation for low-light image enhancement," *International Journal of Computer Vision*, vol. 133, no. 11, pp. 8234–8252, 2025.

[8] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, and Y. Zhang, "Retinexformer: One-stage retinex-based transformer for low-light image enhancement," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 12 504–12 513.

[9] T. Wang, K. Zhang, T. Shen, W. Luo, B. Stenger, and T. Lu, "Ultra-high-definition low-light image enhancement: A benchmark and transformer-based method," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 3, 2023, pp. 2654–2662.

[10] X. Wang, Y. Wang, J. Zhou, J. Liu, Y. Gao, Y. Wang, and J. Zheng, "An unsupervised learning method based on u-net++ for low-light image enhancement," *Signal, Image and Video Processing*, vol. 19, no. 4, p. 282, 2025.

[11] Y. Lin, T. Ye, S. Chen, Z. Fu, Y. Wang, W. Chai, Z. Xing, W. Li, L. Zhu, and X. Ding, "Aglldiff: Guiding diffusion models towards unsupervised training-free real-world low-light image enhancement," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 5, 2025, pp. 5307–5315.

[12] X. Zhang, Z. Xu, H. Tang, C. Gu, W. Chen, and A. El Saddik, "Wakeup-darkness: When multimodal meets unsupervised low-light image enhancement," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 21, no. 3, Feb. 2025. [Online]. Available: https://doi.org/10.1145/3711929

[13] G. Sheng, G. Hu, X. Wang, W. Chen, and J. Jiang, "Low-light image enhancement via clustering contrastive learning for visual recognition," *Pattern Recognition*, vol. 164, p. 111554, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320325002146

[14] J. Guo, J. Ma, Á. F. García-Fernández, Y. Zhang, and H. Liang, "A survey on image enhancement for low-light images," *Heliyon*, vol. 9, no. 4, 2023.

[15] H. Xu, H. Zhang, X. Yi, and J. Ma, "Cretinex: A progressive color-shift aware retinex model for low-light image enhancement," *International Journal of Computer Vision*, vol. 132, no. 9, pp. 3610–3632, 2024.

[16] Z. Zhao, H. Lin, D. Shi, and G. Zhou, "A non-regularization self-supervised retinex approach to low-light image enhancement with parameterized illumination estimation," *Pattern Recognition*, vol. 146, p. 110025, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320323007227

[17] G. Fan, Z. Yao, G.-Y. Chen, J.-N. Su, and M. Gan, "Iniretinex: Rethinking retinex-type low-light image enhancer via initialization perspective," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 3, pp. 2834–2842, Apr. 2025. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/32289

[18] X. Yi, H. Xu, H. Zhang, L. Tang, and J. Ma, "Diff-retinex++: Retinex-driven reinforced diffusion model for low-light image enhancement," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 8, pp. 6823–6841, Aug 2025.

[19] W. Wang, R. Luo, W. Yang, and J. Liu, "Unsupervised illumination adaptation for low-light vision," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 9, pp. 5951–5966, Sep. 2024.

[20] L. Ma, T. Ma, C. Xu, J. Liu, X. Fan, Z. Luo, and R. Liu, "Learning with self-calibrator for fast and robust low-light image enhancement," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 10, pp. 9095–9112, Oct 2025.

[21] H. Xu, J. Ma, Y. Yuan, H. Zhang, X. Tian, and X. Guo, "More than lightening: A self-supervised low-light image enhancement method capable for multiple degradations," *IEEE/CAA Journal of Automatica Sinica*, vol. 11, no. 3, pp. 622–637, March 2024.

[22] W. Wang, H. Yang, J. Fu, and J. Liu, "Zero-reference low-light enhancement via physical quadruple priors," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 26 057–26 066.

[23] C. Xie, L. Fei, H. Tao, Y. Hu, W. Zhou, J. Tian Hoe, W. Hu, and Y.-P. Tan, "Residual quotient learning for zero-reference low-light image enhancement," *IEEE Transactions on Image Processing*, vol. 34, pp. 365–378, 2025.

[24] C. Yu, G. Han, M. Pan, X. Wu, and A. Deng, "Zero-tce: Zero reference tri-curve enhancement for low-light images," *Applied Sciences*, vol. 15, no. 2, 2025. [Online]. Available: https://www.mdpi.com/2076-3417/15/2/701

[25] Z. Zhang, S. Zhao, X. Jin, M. Xu, Y. Yang, S. Yan, and M. Wang, "Noise self-regression: A new learning paradigm to enhance low-light images without task-related data," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 2, pp. 1073–1088, Feb 2025.

[26] C.-Y. Chan, W.-C. Siu, Y.-H. Chan, and H. Anthony Chan, "Anlightendiff: Anchoring diffusion probabilistic model on low light image enhancement," *IEEE Transactions on Image Processing*, vol. 33, pp. 6324–6339, 2024.

[27] X. Guo, Y. Li, and H. Ling, "Lime: Low-light image enhancement via illumination map estimation," *IEEE Transactions on Image Processing*, vol. 26, no. 2, pp. 982–993, Feb 2017.

[28] S. Du, M. Zhao, Y. Liu, Z. You, Z. Shi, J. Li, and Z. Xu, "Low-light image enhancement and denoising via dual-constrained retinex model," *Applied Mathematical Modelling*, vol. 116, pp. 1–15, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0307904X22005637

[29] X. Li, W. Wang, X. Feng, and M. Li, "Deep parametric retinex decomposition model for low-light image enhancement," *Computer Vision and Image Understanding*, vol. 241, p. 103948, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1077314224000298

[30] X. Guo and Q. Hu, "Low-light image enhancement via breaking down the darkness," *International Journal of Computer Vision*, vol. 131, no. 1, pp. 48–66, 2023.

[31] L.-W. Wang, Z.-S. Liu, W.-C. Siu, and D. P. K. Lun, "Lightening network for low-light image enhancement," *IEEE Transactions on Image Processing*, vol. 29, pp. 7984–7996, 2020.

[32] C. Li, C. Guo, and C. C. Loy, "Learning to enhance low-light image via zero-reference deep curve estimation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 4225–4238, Aug 2022.

[33] X. Lyu and J. Hou, "Enhancing low-light light field images with a deep compensation unfolding network," *IEEE Transactions on Image Processing*, vol. 33, pp. 4131–4144, 2024.

[34] Q. Yang, Q. Cheng, H. Yue, L. Zhang, Y. Liu, and J. Yang, "Learning to see low-light images via feature domain adaptation," *IEEE Transactions on Image Processing*, vol. 34, pp. 2680–2693, 2025.

[35] A. Razafindratovolahy, Y. Rao, S. Zeng, F. L. Delali, J. Xue, and C. Sey, "Smile: Enhancing low-light images in lightweight networks

via exposure-aware non-reference losses," *Multimedia Systems*, 2025, accepted for publication.

[36] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Physica D: Nonlinear Phenomena*, vol. 60, no. 1, pp. 259–268, 1992. [Online]. Available: https://www.sciencedirect.com/science/article/pii/016727899290242F

[37] G. Buchsbaum, "A spatial processor model for object colour perception," *Journal of the Franklin Institute*, vol. 310, no. 1, pp. 1–26, 1980. [Online]. Available: https://www.sciencedirect.com/science/article/pii/0016003280900587

[38] C. Wei, W. Wang, W. Yang, and J. Liu, "Deep retinex decomposition for low-light enhancement," *arXiv preprint arXiv:1808.04560*, 2018.

[39] W. Yang, W. Wang, H. Huang, S. Wang, and J. Liu, "Sparse gradient regularized deep retinex network for robust low-light image enhancement," *IEEE Transactions on Image Processing*, vol. 30, pp. 2072–2086, 2021.

[40] C. Lee, C. Lee, and C.-S. Kim, "Contrast enhancement based on layered difference representation," in *2012 19th IEEE international conference on image processing*. IEEE, 2012, pp. 965–968.

[41] X. Guo, Y. Li, and H. Ling, "Lime: Low-light image enhancement via illumination map estimation," *IEEE Transactions on image processing*, vol. 26, no. 2, pp. 982–993, 2016.

[42] K. Ma, K. Zeng, and Z. Wang, "Perceptual quality assessment for multi-exposure image fusion," *IEEE Transactions on Image Processing*, vol. 24, no. 11, pp. 3345–3356, 2015.

[43] S. Wang, J. Zheng, H.-M. Hu, and B. Li, "Naturalness preserved enhancement algorithm for non-uniform illumination images," *IEEE transactions on image processing*, vol. 22, no. 9, pp. 3538–3548, 2013.

[44] A. Razafindratovolahy and Y. Rao, "A simple yet robust nonlinear function for low-light image enhancement task," *IEEE Signal Processing Letters*, vol. 32, pp. 3370–3374, 2025.

[45] A. Razafindratovolahy, Y. Rao, J. C. Tovolahy, M. Afanga, and A. Mutale, "Tovo: Tone-oriented vision optimization for efficient low-light image enhancement," *Signal Processing*, vol. 244, p. 110540, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S016516842600054X

[46] Q. Zhang, Y. Nie, and W. Zheng, "Dual illumination estimation for robust exposure correction," *Computer Graphics Forum*, vol. 38, pp. 243–252, 11 2019.

[47] W. Chen, W. Wang, W. Yang, and J. Liu, "Deep retinex decomposition for low-light enhancement," in *Proceedings of the British Machine Vision Conference*. British Machine Vision Association, 2018.

[48] Y. Zhang, X. Guo, J. Ma, W. Liu, and J. Zhang, "Beyond brightening low-light images," *International Journal of Computer Vision*, vol. 129, pp. 1013–1037, 2021.

[49] Y. Cai, H. Bian, J. Lin, H. Wang, R. Timofte, and Y. Zhang, "Retinexformer: One-stage retinex-based transformer for low-light image enhancement," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 12 470–12 479.

[50] Z. Fu, Y. Yang, X. Tu, Y. Huang, X. Ding, and K.-K. Ma, "Learning a simple low-light image enhancer from paired low-light instances," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 22 252–22 261.

[51] H. Zhang, L. Wang, Q. Zou, and J. Zeng, "Dff-net: Deep feature fusion network for low-light image enhancement," *Image and Vision Computing*, vol. 161, p. 105645, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0262885625002331

[52] Y. Wang, L. Jiang, Z. Du, B. Li, and W. Yang, "Csbnet: Leveraging edge intelligence for multigranularity low-light image enhancement," *IEEE Internet of Things Journal*, vol. 12, no. 8, pp. 10 558–10 573, 2025.

[53] Y. Jiang, X. Gong, D. Liu, Y. Cheng, C. Fang, X. Shen, J. Yang, P. Zhou, and Z. Wang, "Enlightengan: Deep light enhancement without paired supervision," *IEEE transactions on image processing*, vol. 30, pp. 2340–2349, 2021.

Biography Section

Annicet Razafindratovolahy received his BS in Information and Communication Technology and ME in Electronics and Industrial Computing from Ecole Supérieure Polytechnique d'Antsiranana, Madagascar, in 2016 and 2019, respectively. He is pursuing his Ph.D. in Software Engineering at the University of Electronic Science and Technology of China. His research interests include image/video processing, with a focus on visual data enhancement and reliable data quality improvement. He is also interested in visual understanding tasks such as recognition and scene understanding, as well as vision-language models.

Yunbo Rao (M'19) received the BS degree from Sichuan Normal University, in 2003, and the ME and Ph.D. degrees from the School of Computer Science and Engineering, University of Electronic Science and Technology of China, in 2006 and 2012, respectively. From 2009 to 2011, he was a Visiting Scholar of Electrical Engineering at the University of Washington, Seattle, USA. Since 2012, he has been with the School of Information and Software Engineering, University of Electronic Science and Technology of China, where he is currently an Associate Professor. His research interests include image segmentation, 3-D reconstruction, video enhancement, and medical image processing.

Shaoning Zeng (M'18) received his B.S. and M.S. from Beihang University, Beijing, China, in 2004 and 2007, respectively. He received the Ph.D. degree in Computer Science from the University of Macau in 2020. Currently, he is an Associate Professor in the Yangtze Delta Region Institute (Huzhou) of University of Electronic Science and Technology of China. His research focus includes but is not limited to: image segmentation, 3-D reconstruction, video and image enhancement, and medical image processing.

Mona Afanga received her DEUG degree from the School of Mathematical Science and Informatics and her BS in Software Engineering from Ibn Zohr University in 2016 and 2017, respectively. She received her ME from the School of Information and Software Engineering at the University of Electronic Science and Technology of China in 2022. She is currently a Ph.D. candidate at the School of Computer Science and Engineering at the University of Electronic Science and Technology of China. Her research interests include, but are not limited to, object recognition, image and text processing, and 3D generation.

Ssemwogerere Rajab (he is Rajab M. S. Currently pursing a PhD. Software Engineering highly motivated to engage in advanced research and innovation across computer-aided diagnosis, computer vision, artificial intelligence, pattern recognition, and large language models. I strive to build a strong professional network with leading researchers & practitioners in these interdisciplinary fields. Through collaboration, I aim to contribute to the development of state-of-the-art techniques and impactful applications that push the boundaries of computer science & medical imaging.

Todizara Andrianajaina is a project manager and researcher with a PhD in Science and Technology focused on renewable energy and artificial intelligence. With over seven years of experience, his work spans educational, agricultural, and digital innovation projects. He combines academic expertise with practical field experience to promote youth empowerment, sustainable development, and community resilience. Formerly a university career advisor and now a project leader at the SOA Network, he is dedicated to bridging research, AI, and local initiatives to generate lasting social and environmental impact.